

Negative binomial regression

Last time: class activity

$$\beta_4 - \beta_3 \quad : \quad \text{Contrast}$$

$$1\beta_4 - 1\beta_3$$

$$1 - 1 = 0$$

(a linear combination where the coefficients sum to 0)

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	2.837192	0.115099	24.650	< 2e-16 ***
male	0.456170	0.026095	17.481	< 2e-16 ***
age	-0.006806	0.001595	-4.267	2.07e-05 ***
education2	0.016463	0.029675	0.555	0.579
education3	0.016427	0.037706	0.436	0.663
education4	-0.015033	0.040477	-0.371	0.710
diabetes	-0.025398	0.092606	-0.274	0.784
BMI	0.005001	0.003304	1.513	0.130

Is there a difference in the average number of cigarettes smoked between smokers with some college education and smokers with a college degree, after accounting for sex, age, diabetes, and BMI?

$$H_0: \beta_3 = \beta_4 \Rightarrow \beta_4 - \beta_3 = 0$$

$$H_A: \beta_3 \neq \beta_4 \rightarrow \beta_4 - \beta_3 \neq 0$$

$$t = \frac{(\hat{\beta}_4 - \hat{\beta}_3) - 0}{SE(\hat{\beta}_4 - \hat{\beta}_3)}$$

var in R: variance-covariance matrix

Last time: class activity

random variables

$$Var(aU + bV) = a^2 Var(U) + b^2 Var(V) + 2ab Cov(U, V)$$

constants

$$SE(\hat{\beta}_4 - \hat{\beta}_3) = \sqrt{Var(\hat{\beta}_4 - \hat{\beta}_3)}$$

$$Var(\hat{\beta}_4 - \hat{\beta}_3) = Var(1\hat{\beta}_4 - 1\hat{\beta}_3)$$

$$= 1^2 Var(\hat{\beta}_4) + (-1)^2 Var(\hat{\beta}_3)$$

$$+ 2(1)(-1) Cov(\hat{\beta}_4, \hat{\beta}_3)$$

$$= \underset{\substack{\uparrow \\ 0.00142}}{Var(\hat{\beta}_4)} + \underset{\substack{\uparrow \\ 0.000881}}{Var(\hat{\beta}_3)} - 2 \underbrace{Cov(\hat{\beta}_4, \hat{\beta}_3)}_{0.00042}$$

$$\begin{bmatrix} Var(\hat{\beta}_0) & Cov(\hat{\beta}_0, \hat{\beta}_1) \\ Cov(\hat{\beta}_0, \hat{\beta}_1) & Var(\hat{\beta}_1) \\ & & Var(\hat{\beta}_2) \\ & & & \ddots \end{bmatrix}$$

$$\begin{aligned} a &= 1 \\ b &= -1 \\ u &= \hat{\beta}_4 \\ v &= \hat{\beta}_3 \end{aligned}$$

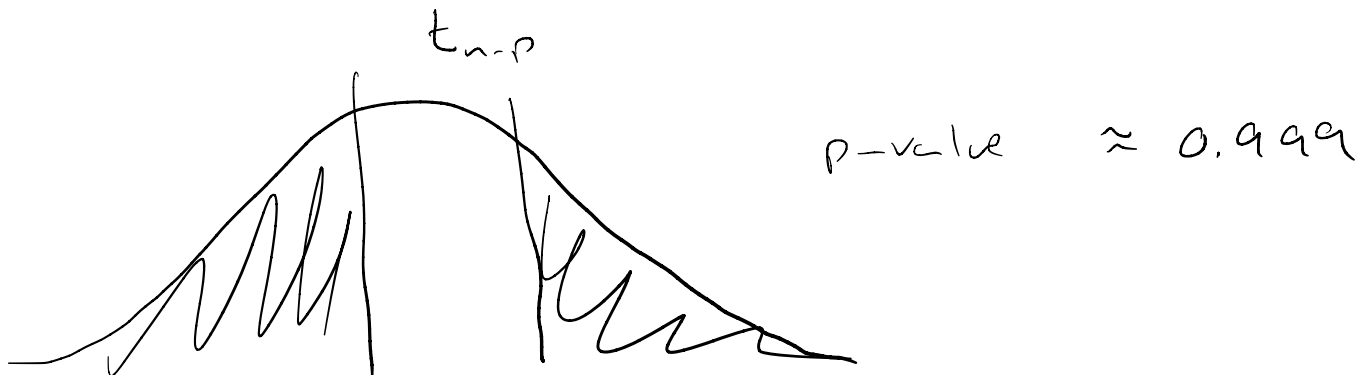
Last time: class activity

```
se <- sqrt(1.421765e-03 + 8.806037e-04 - 2*4.199471e-04) # 0.0382
t_stat <- (0.016427 - 0.016463)/se # -0.001
2*pt(abs(t_stat), 2004, lower.tail=F)
```

```
## [1] 0.999249
```

$$SE(\hat{\beta}_4 - \hat{\beta}_3) = \sqrt{\text{var}(\hat{\beta}_4 - \hat{\beta}_3)} = 0.0382$$

$$t = \frac{0.016427 - 0.016463}{0.0382} \approx -0.001$$



Comparing Poisson and quasi-Poisson

Poisson:

- + Mean = λ_i
- + Variance = λ_i

quasi-Poisson:

- + Mean = λ_i
 - + Variance = $\phi \lambda_i$
 - + Variance is a linear function of the mean
- ϕ : dispersion parameter

What if we want variance to depend on the mean in a different way?

Introducing the negative binomial

If $Y_i \sim NB(\theta, p)$, then Y_i takes values $y = 0, 1, 2, 3, \dots$ with probabilities

$$P(Y_i = y) = \frac{(y + \theta - 1)!}{y!(\theta - 1)!} (1 - p)^\theta p^y$$

↙ size parameter in $\mathcal{R}$
+ $\theta > 0, \quad p \in [0, 1]$

+ Mean = $\frac{p\theta}{1 - p} = \mu$

+ Variance = $\frac{p\theta}{(1 - p)^2} = \mu + \frac{\mu^2}{\theta}$

+ Variance is a *quadratic* function of the mean

Mean and variance for a negative binomial variable

As $\theta \rightarrow \infty$ (μ fixed)

$NB(\theta, p) \rightarrow \text{Poisson}(\mu)$

If $Y_i \sim NB(\theta, p)$, then

+ Mean = $\frac{p\theta}{1-p} = \mu$

+ Variance = $\frac{p\theta}{(1-p)^2} = \mu + \frac{\mu^2}{\theta}$

How is θ related to overdispersion?

θ small:

$\mu + \frac{\mu^2}{\theta} \gg \mu$
(more overdispersion from Poisson perspective)

θ large:

$\mu + \frac{\mu^2}{\theta} \approx \mu$
(less overdispersion from Poisson perspective)

Negative binomial regression

$$Y_i \sim NB(\theta, p_i)$$

(random component is different than in Poisson regression)

$$\log(\mu_i) = \beta_0 + \beta_1 X_i$$

(systematic component is the same)

$$+ \mu_i = \frac{p_i \theta}{1 - p_i}$$

+ Note that θ is the same for all i

+ Note that just like in Poisson regression, we model the average count

+ Interpretation of β s is the same as in Poisson regression

Comparing Poisson, quasi-Poisson, negative binomial

Poisson:

- + Mean = λ_i
- + Variance = λ_i

quasi-Poisson:

- + Mean = λ_i
- + Variance = $\phi\lambda_i$

negative binomial:

- + Mean = μ_i
- + Variance = $\mu_i + \frac{\mu_i^2}{\theta}$

In R

library(MASS)

m3 <- **glm.nb**(cigsPerDay ~ male + age + education +
diabetes + BMI, data = smokers)

has glm.nb function

don't have to
specify a family

not glm

	Estimate	Std. Error	(z) value	Pr(> z)	
(Intercept)	2.877771	0.123477	23.306	< 2e-16	***
male	0.459148	0.027641	16.611	< 2e-16	***
age	-0.007010	0.001731	-4.050	5.12e-05	***
education2	0.024518	0.032534	0.754	0.451	
education3	0.009252	0.040802	0.227	0.821	
education4	-0.027732	0.044825	-0.619	0.536	
diabetes	-0.010124	0.099126	-0.102	0.919	
BMI	0.003693	0.003573	1.033	0.301	

(Dispersion parameter for Negative Binomial(3.2981) family

$\hat{\theta}$

$$\hat{\theta} = 3.30$$

In R

$$\log(\mu_i) = \dots$$

$$e^{-0.007} \approx 0.993 \Rightarrow \text{decrease of } (1 - 0.993) \times 100\%$$

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	2.8777771	0.123477	23.306	< 2e-16	***
male	0.459148	0.027641	16.611	< 2e-16	***
age	-0.007010	0.001731	-4.050	5.12e-05	***
education2	0.024518	0.032534	0.754	0.451	
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diabetes	-0.010124	0.099126	-0.102	0.919	
BMI	0.003693	0.003573	1.033	0.301	

How do I interpret the estimated coefficient -0.007?

Same as for Poisson & quasi-Poisson:

A one-year increase in age is associated w/
a decrease in the mean # of cigarettes smoked
by a factor of $e^{-0.007}$ (0.7% decrease),
holding other variables constant

Hypothesis testing

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	2.877771	0.123477	23.306	< 2e-16	***
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age	-0.007010	0.001731	-4.050	5.12e-05	***
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diabetes	-0.010124	0.099126	-0.102	0.919	
BMI	0.003693	0.003573	1.033	0.301	

Research question: Is there a relationship between education level and the number of cigarettes smoked per day, after accounting for sex, age, diabetes, and BMI?

$$H_0: \beta_3 = \beta_4 = \beta_5 = 0$$

$$H_A: \text{at least one of } \beta_3, \beta_4, \beta_5 \neq 0$$

Poisson regression ;
(or logistic)

$$G = \text{deviance}_{\text{reduced}} - \text{deviance}_{\text{full}}$$

$$\text{deviance} = -2 \log L + c$$

$$= 2 \log L_{\text{full}} - 2 \log L_{\text{reduced}}$$

$$= 2 \log \left(\frac{L_{\text{full}}}{L_{\text{reduced}}} \right)$$

$\Rightarrow$ Likelihood ratio test

NB model ;

$$G = 2 \log L_{\text{full}} - 2 \log L_{\text{reduced}}$$

need to calculate log-likelihood directly
(don't use deviance for NB)

Likelihood ratio test

```
m3 <- glm.nb(cigsPerDay ~ male + age + education +  
              diabetes + BMI, data = smokers)  
m4 <- glm.nb(cigsPerDay ~ male + age +  
              diabetes + BMI, data = smokers)  
m3$twologlik - m4$twologlik
```

↙ full model

↙ reduced model

```
## [1] 1.423055
```

```
pchisq(1.423, df=3, lower.tail=F)
```

```
## [1] 0.7001524
```

Under H_0 : $G \sim \chi^2_g$

$g = \#$ parameters tested

quasi-Poisson vs. negative binomial

quasi-Poisson:

- + linear relationship between mean and variance
- + easy to interpret $\hat{\phi}$
- + same as Poisson regression when $\phi = 1$
- + simple adjustment to estimated standard errors
- + estimated coefficients same as in Poisson regression

negative binomial:

- + quadratic relationship between mean and variance
- + we get to use a likelihood, rather than a quasi-likelihood
- + Same as Poisson regression when θ is very large and p is very small

Class activity

Work on the class activity. Submit the rendered HTML on Canvas at the end of class.

Class activity

```
m_art <- glm.nb(art ~ ., data = articles)
```

	Estimate	Std. Error	z value	Pr(> z)	
(Intercept)	0.256144	0.137348	1.865	0.062191	.
femWomen	-0.216418	0.072636	-2.979	0.002887	**
marMarried	0.150489	0.082097	1.833	0.066791	.
kid5	-0.176415	0.052813	-3.340	0.000837	***
phd	0.015271	0.035873	0.426	0.670326	
ment	0.029082	0.003214	9.048	< 2e-16	***

Class activity

```
m_art <- glm.nb(art ~ ., data = articles)
m_art_reduced <- glm.nb(art ~ fem + mar +
                        kid5 + ment,
                        data = articles)
m_art$twologlik - m_art_reduced$twologlik
```

```
## [1] 0.1795491
```

```
pchisq(0.180, df=1, lower.tail=F)
```

```
## [1] 0.6713732
```